

## UNIT - 2

### 1) Threats and Attacks:

\* Information Security threats can be many like software attacks, theft of intellectual property, identity theft, theft of equipment (or) information, sabotage and information extortion.

\* Threat can be anything that can take advantage of a vulnerability to breach security and negatively alter, erase, harm object (or) objects of interest.

\* Software attacks means attack by viruses, worms, Trojan Horses etc.

\* Many users believe that malware, virus, worms, bots are all same things But they are not same, only similarity is that they all are malicious software that behaves differently.

\* Malware is a combination of 2 terms malicious and software. So malware basically means malicious software that can be an intrusive program (or) anything that is designed to perform malicious operations on system.

\* Malware can be divided into two.

→ Infection Methods

→ Malware Actions.

\* Malware on the basis of Infection Methods are following.

\* Virus: The Ability to replicate themselves by hooking them to the program on the host

Computer like songs, videos etc.

- \* Worms : Worms are also self-replicative in nature but they don't look themselves.

- \* Trojan : Trojan is defined from the "Trojan Horse" tale in Greek mythology.

- \* Bots : Advanced form of worms. They are automated processes that are designed to interact over the internet without need for human interaction.

- \* Adware : Not exactly malicious but they do breach privacy of the users. They display Ads on a computer's display (or) inside installed programs.

- \* Spyware : It is a program (or) we can say software that monitors your activities on computers.

- \* Random Ware : It is type of Malware that will either encrypt your files (or) will lock your computer making it inaccessible.

- \* Zombies : They work similar to spyware. Infection mechanism is same but don't spy and steal information rather they will wait for the command from hackers.

- \* Theft of Intellectual property violation of Intellectual property like copyrights, patents etc.

- \* Identity theft obtain person's information (or) to access vital information they have like accessing vital accessing the computer.

- \* Theft of equipment and information mobile nature of devices are increasing information capacity.

- \* Sabotage destroying company's property (or) information to receive payment in exchange.
- \* Information Extortion destroying company's website to cause loss of confidence on parties of its new customers.

\* New generation threats :

- \* Technology with weak security
- \* Social media attacks
- \* Mobile Malware
- \* outdated security software
- \* corporate data on personal devices
- \* Social engineering.

## Attacks:

- \* Attacks are defined as passive and Active.
- \* A passive attack is an attempt to understand (or) create use of data from the system without influencing system resource.
- \* Whereas an active attack is an attempt to change system resources (or) influence their operation.
- \* Active attacks
- \* Passive attacks
- \* Active attacks:
  - \* An active attacks attempts to alter system resources (or) affect their operations.
  - \* Active attacks involve some modifications of the data stream (or) the creation of false statements. types of attacks are
    - \* Masquerade

\* Modification of messages.

\* Repudiation

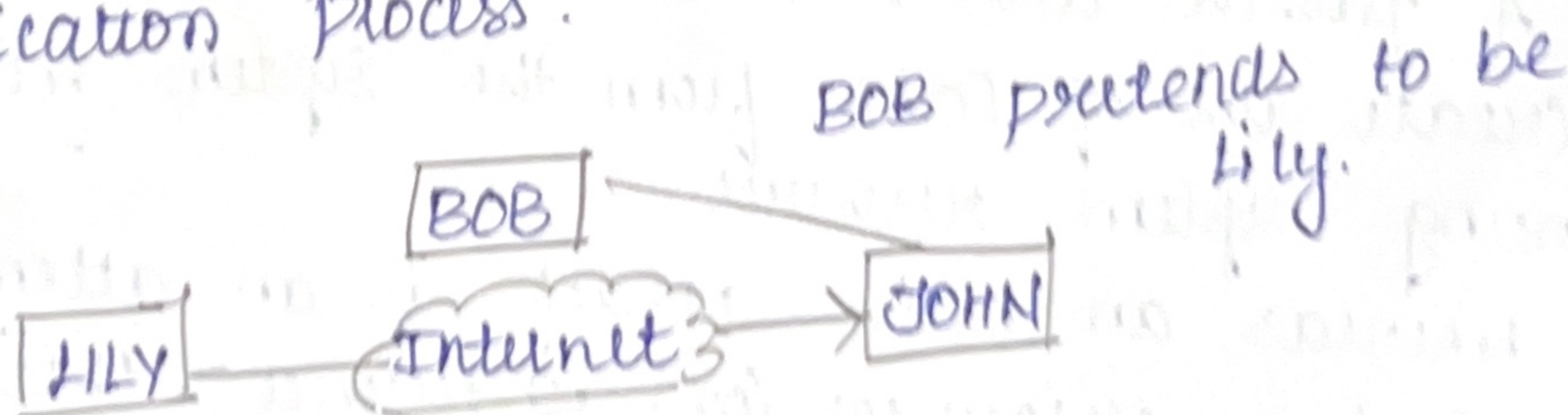
\* Replay

\* Denial of service.

Masquerade :

\* A Masquerade Attack takes place when one entity pretends to be a different entity. A masquerade attack involves one of the other forms of active attacks.

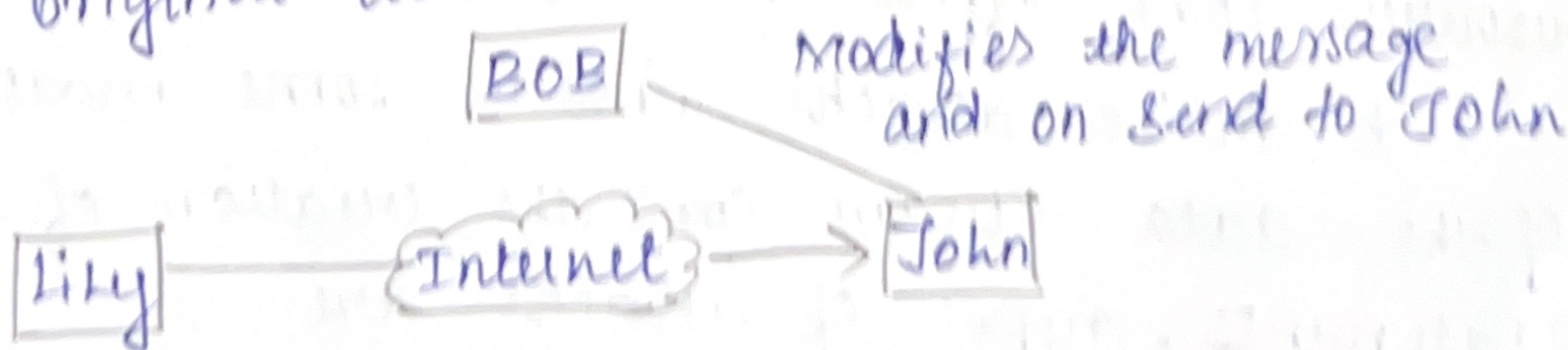
\* If an authorization procedure isn't always absolutely protected, it is able to grow to be extraordinarily assaults may be performed using the stolen passwords and logins, with the aid of using finding gaps in programs (or) with the aid of using locating a manner across the authentication process.



Modification of message:

\* It means that some portion of a message is altered (or) that message is delayed (or) recorded to produce an unauthorized effect.

\* Modification is an attack on the integrity of the original data.



## Repudiation :

\* This attack occurs when the network is not completely secured (or) the login control has been tampered with. With this attack, the author's info can be changed by actions of a malicious user in order to save false data in log files, up to the general manipulation of data on behalf of others, similar to the spoofing of e-mail message.

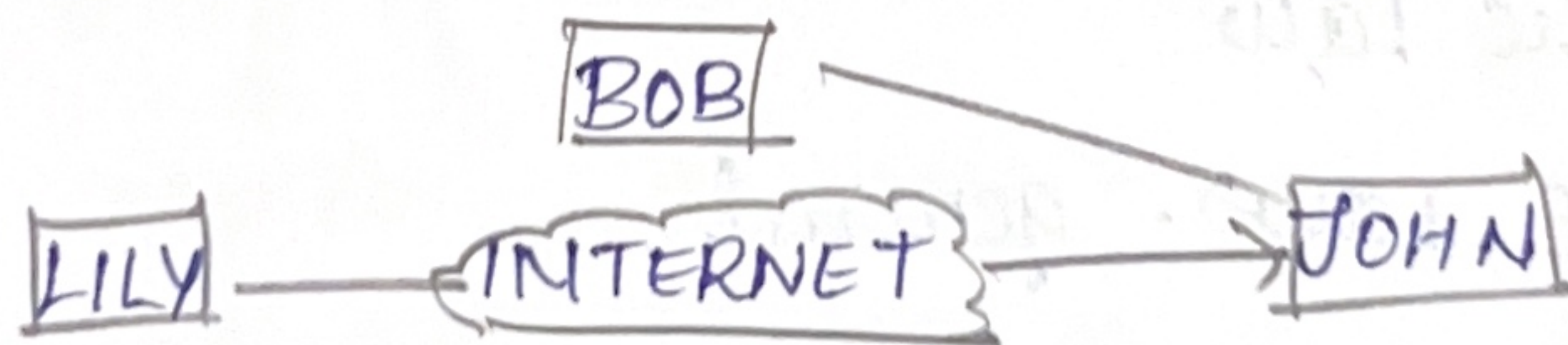
## Replay:

\* It involves the passive capture of a message and its subsequent transmission to produce an authorized effect.

\* Once the data is corrupted (or) leaked it is insecure and unsafe for the users.



## Denial of Service :



\* It prevents the normal use of communications facilities.

\* For example, an entity may suppress all messages directed to a particular destination.

## Passive Attack :

\*) A passive attack attempts to learn (or) make use of information from the system but does not affect system resources.

\*) Types of passive attacks are

- \* The Release of message content

- \* Traffic Analysis

## 2) Legal Ethical and Professional Issues:

- \* Law and Ethics in Information Security.

- \* Laws are rules that mandate (or) prohibit certain behaviour in society, they are drawn from ethics, which define socially acceptable behaviours.

- \* The key difference between laws and ethics is that Laws carry the sanctions of a governing authority and ethics do not.

- \* Ethics in turn are based on cultural mores.

### Types of Law:

- \* Civil Law

- \* Criminal Law

- \* Tort Law

- \* Private Law

- \* Public Law

- \* Relevant U.S. Laws - general.

- \* Computer Fraud and Abuse Act of 1986 (CFAA)

- \* National Information Infrastructure Protection Act of 1996.

- \* USA Patriot Act of 2001.

- \* Telecommunications Deregulation and competition Act of 1996.

- \* Communications Decency Act of 1996 (CDA)

- \* Computer Security Act of 1987.

\* You must understand Scope of an organization's legal and ethical responsibilities.

\* To minimize liabilities / reduce risks, the information security practitioners must.

\* ) Understand current legal environment.

\* ) Stay current with laws and regulations.

\* ) Watch for new issues that emerge.

\* ) Privacy:

\* One of the hottest topic in information security.

\* Is a "state of being free from unsanctioned intrusive.

\* Ability to aggregate data from multiple sources allow creation of information databases previously.

\* ) Privacy of Customer Information :-

\* Privacy of customer information section of common carrier regulations.

\* Federal Privacy Act of 1974. The electronic communications privacy Act of 1986.

\* The health Insurance portability Accountability Act of 1996 (HIPAA) also known as the Kennedy-Kassebaum Act.

\* The Financial Services Modernization Act (or) Gramm-Leach-Bliley Act of 1999.